<table>
<tr><th colspan="3" rowspan="2">Question</th><th rowspan="2">Marking details</th><th></th><th colspan="3">Marks available</th><th></th><th></th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>4</td><td>(a)</td><td>(i)</td><td>
<table>
<tr><th>Particle</th><th>Symbol</th><th>Quark Combination</th><th>Charge/e</th><th>Baryon Number</th></tr>
<tr><td>Proton</td><td>p</td><td>uud</td><td>+1</td><td>1</td></tr>
<tr><td>Delta particle</td><td>Δ^{++}</td><td>uuu</td><td>+2</td><td>1</td></tr>
<tr><td>Electron</td><td>e^-</td><td>No quarks present</td><td>-1</td><td>0</td></tr>
<tr><td>Pion</td><td>π^-</td><td>$\bar{u}d$ or $d\bar{u}$</td><td>-1</td><td>0</td></tr>
</table>
[3 × 1] marks for each correct row (ignoring row for proton)</td><td></td><td>3</td><td></td><td>3</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Electron</td><td>1</td><td></td><td></td><td>1</td><td></td><td></td></tr>
<tr><td></td><td>(b)</td><td></td><td>Charge:
LHS: -1 +1 (=0)
RHS: -1 +2 -1 (=0) [must be shown how 0 is determined on both sides] (1)
Lepton Number:
LHS: 1 + 0 (= 1)
RHS: 1 + 0 + 0 (=1) [must be shown how 1 is determined on both sides] (1)</td><td></td><td>2</td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td>(c)</td><td>(i)</td><td>Up Quark: Δ^{++} contains 3 up quarks (uuu) = proton contains 2 up quarks (uud) + pion contains 1 up quark (ud) (1)
Down Quark: Δ^{++} contains 0 down quarks = proton contains 1down quark (uud) + pion contains 1 antidown quark (ud). (1)
Alternative for first mark: equation in quark form $uuu \rightarrow uud + u\bar{d}$</td><td></td><td>2</td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Any 2 × (1) from:
• Very short lifetime / decays quickly / ref to 6×10^{-24} s
• No change in u or d quark number (or flavour)
• Only quarks or hadrons involved
• No γ (or photon) or no neutrino involvement</td><td>2</td><td></td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td>(d)</td><td></td><td>No practical use for electron or proton or both when they were discovered (1)
Now many uses and one example given for either electron or proton (1)</td><td></td><td></td><td>2</td><td>2</td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td>Question 4 total</td><td>3</td><td>7</td><td>2</td><td>12</td><td>0</td><td>0</td></tr>
</table>